Xebia

Agentic Coding Beyond Autocomplete

Real Work with Copilot Agent Mode and Claude Code

Eric Greene · Xebia

eric.greene@xebia.com

Agenda

- From inline suggestions to autonomous agents
- How Copilot agent mode works
- How Claude Code works
- Patterns: refactors, tests, debugging, and new features
- Keeping agents on track: prompts, intervention, and review
- Team adoption: repo hygiene, context, and workflow

From suggestions to agents

Autocomplete

Finishes the line you're typing

You drive every keystroke

Chat

Answers questions, drafts code

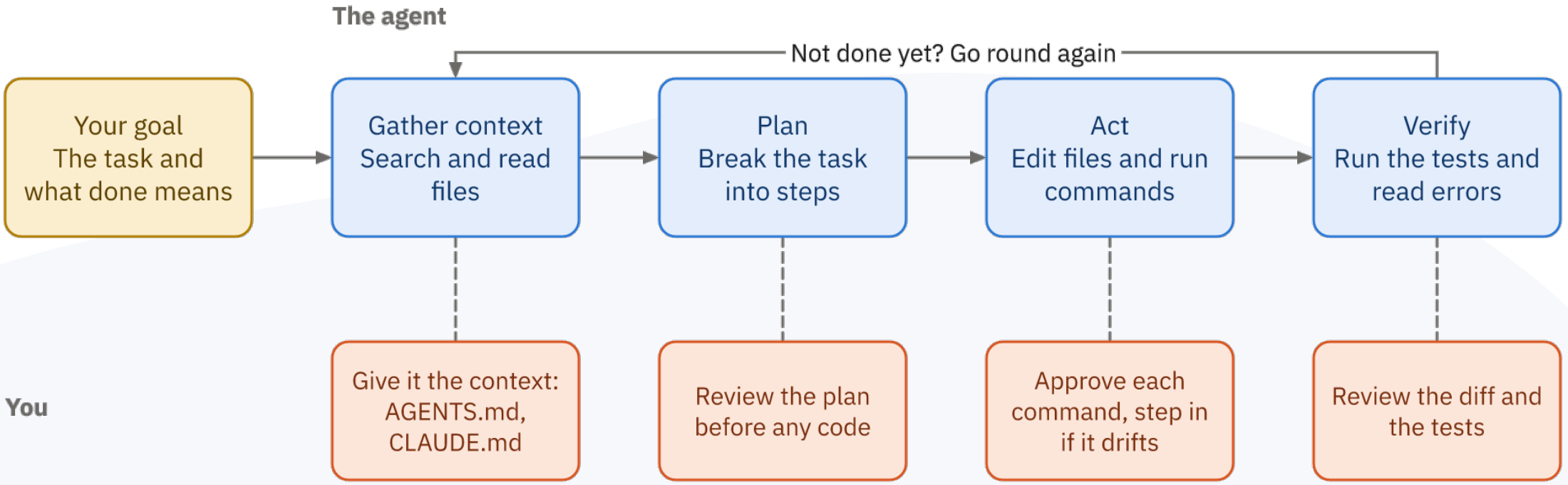
You copy, paste, and run it

Agent

Plans, edits, runs, and iterates

You set the goal and review

The agent loop



What changes, and what breaks

What changes

Scope grows from a line to a codebase

The agent runs commands, not just you

Your job shifts to direction and review

What breaks

Review load grows with every file touched

Confident changes built on missing context

Tests quietly edited until they pass

How Copilot agent mode works

- Ask, Plan, and Agent, in the Copilot harness
- Searches the workspace and reads files itself
- Edits several files, reviewed as one diff
- Runs commands after you select Allow
- Reads test output and errors, then tries again
- Checkpoints roll the workspace back to any request

Demo 1: From autocomplete to agent

demos/01-copilot-agent-mode · about 17 minutes

How Claude Code works

- Gathers context with search and read tools
- Loads CLAUDE.md into every session
- Edits files and runs commands as tool calls
- Every tool call passes a permission check
- Plan mode thinks before it changes anything
- Rewind undoes edits you don't want

Approvals and permission modes

Manual

Edits and most commands ask

For unfamiliar or risky work

Edit automatically

Edits flow, most commands ask

When tests check every change

Plan

Reads and plans, changes nothing

Before a change worth reviewing

A CLAUDE.md worth loading

```
# Roombook
## Commands
- Run the tests: `uv run pytest`
- Run the API: `uv run uvicorn roombook.main:app`
## How it fits together
- Times are stored in UTC, sent in office time
- Every booking goes through policies.py
## Rules
- Don't edit a test to make code pass
- Ask before adding a dependency
```

Demo 2: Context, tools, and approvals

demos/02-claude-code · about 17 minutes

Refactors and test generation

Refactors

Plan first, and read it for behavior changes

Tests pass before and after, unchanged

Structure or behavior, never both at once

Test generation

Expected values from real runs or the spec

Pin the edges: limits, boundaries, the clock

Coverage finds gaps; it isn't the goal

Debugging and new features

Debugging sessions

Reproduce with a failing test first

Ask for the root cause and evidence

Ask what else uses the faulty code

New features

Start from a short written spec

Review the plan before any code

Build in slices, with tests in each

Demo 3: Debug and refactor with agents

demos/03-debug-and-refactor · about 18 minutes

A prompt with structure

```
Goal:        Weekly recurring bookings, so teams
             stop booking the same room by hand.
Context:     Reuse check_slot in roombook/slots.py.
Constraints: No new dependencies.
             Don't change the existing tests.
Done when:   A weekly 10:00 booking stays at 10:00
             after New York's clocks change,
             and every test passes.
```

When to step in, how to review

Step in when the agent

Edits a test to make it pass

Adds a dependency or touches unrelated files

Repeats the same failing step

Review the output by

Running the tests yourself

Reading every diff, tests included

Asking a second agent, then deciding

Fitting agents into a team

Repo hygiene

Clear commands to test and run

Fast tests an agent can trust

Context quality

One AGENTS.md for every tool

Lessons learned go back into it

Workflow fit

Shared rules and hooks in git

Same review for agent changes

Demo 4: Ship a feature, keep it on track

demos/04-team-feature · about 18 minutes

Next steps

- Clone the repo and run the four demos
- Try one real task in each tool this week
- Write a short AGENTS.md for one of your repos
- Check in shared permissions and a formatting hook
- Start from the repo's starter kit and checklist



Thank you

Eric Greene · Xebia

eric.greene@xebia.com

xebia.com